Asset	STRIDE (Threat) / Safety Risk	Threat Scenario (EN)	Requirement (shall) (EN)	Evidence (Representative Docs)
DF.1.1.1 Prompt (query/command/requests), Authentication(Session, ID), File Data	Tampering	Direct/indirect prompt injection into inputs or context contaminates system instructions and bypasses safety policies.	The system must implement prompt-injection detection/blocking for all inputs and contexts and preserve system instructions.	MITRE ATLAS, OWASP Top 10 for LLM Apps, ISO/IEC 23894, ENISA Threat Landscape 2023, OWASP ASVS, CWE-78/94
	Info Disclosure	Injection or conversation-flow manipulation reveals sensitive data or the system prompt.	The system must implement prompt-injection detection/blocking for all inputs and contexts and preserve system instructions.	ISO/IEC 27701, GDPR, MITRE ATLAS, OWASP LLM Top 10
	DoS	Repeated/parallel/looped requests trigger compute overload and billing spikes.	The system must enforce rate limits and caps on retries/concurrency per session/user/API key to prevent resource exhaustion and excessive billing.	OWASP API Security Top 10 (2023), OWASP LLM Top 10, ENISA TL 2023
	Spoofing	API keys, tokens, or session IDs are stolen from input paths or active sessions.	Store credentials in a secure vault, encrypt in transit and at rest, and immediately revoke/rotate if exposure is detected.	NIST SP 800-63B, MITRE ATLAS, OWASP ASVS
	Repudiation	Missing or tampered request/response/config logs weaken non-repudiation.	Maintain tamper-evident audit logs with central collection, signing, and integrity verification; control access and changes.	NIST SP 800-92, OWASP ASVS
	Elevation of Privilege	Abusing upload or serialization paths enables remote code execution.	Avoid unsafe (de)serialization and enforce schema validation, timeouts, and size limits.	CWE-502, OWASP Top 10 (Web) 2021 A8, MITRE ATT&CK/ATLAS
DF.2.1.1 File Data/Response(Text/Code)	Tampering	Stealth code, links, or QR payloads are embedded in generated outputs or attachments.	Allow only safe markup for rendering outputs and statically analyze to block injected code, links, or QR content.	OWASP LLM Top 10, CWE-79/94, ISO/IEC 23894
	Info Disclosure	Sensitive data leaks via HTTP/web responses.	Apply DLP, masking, and blocking policies so sensitive data is not included in outputs, logs, responses, or external transmissions.	ISO/IEC 27701, OWASP API Security Top 10, ENISA TL
	DoS	Mass crawling or request bursts cause latency and outages.	Apply rate-limiting, backoff, and bot-blocking to crawling and bulk requests to prevent outages.	OWASP API Security Top 10, ENISA TL
	Elevation of Privilege	Handling a response with a deserialization flaw leads to remote code execution.	Avoid unsafe (de)serialization and enforce schema validation, timeouts, and size limits.	CWE-502, OWASP Top 10 (Web) 2021 A8
DF.1.2.1 Authentication Metadata	Spoofing	Credentials or session IDs are stolen from metadata, environment variables, or sessions.	Store credentials in a secure vault, encrypt in transit and at rest, and immediately revoke/rotate if exposure is detected.	NIST SP 800-63B, OWASP ASVS, MITRE ATLAS
	Info Disclosure	Authentication endpoints or schemas are exposed.	Protect auth endpoints, schemas, and traffic with mTLS/signature verification and keep documentation private.	OWASP API Security Top 10, ASVS
	Tampering	Auth-related API traffic is intercepted or tampered with.	Protect auth endpoints, schemas, and traffic with mTLS/signature verification and keep documentation private.	OWASP Web Security, ATT&CK/ATLAS
DF.2.2.1 Task Automation Responses	Tampering	Auto-responses execute and propagate malicious workflow commands.	Minimize workflow/agent privileges, and execute high-risk commands only after human-in-the-loop approval.	OWASP LLM Top 10, MITRE ATLAS
	DoS	Automation agents loop or issue excessive calls, causing overload.	The system must enforce rate limits and caps on retries/concurrency per session/user/API key to prevent resource exhaustion and excessive billing.	ENISA TL, OWASP LLM Top 10
	Info Disclosure	AI usage status or model versions appear in responses or logs.	Maintain tamper-evident audit logs with central collection, signing, and integrity verification; control access and changes.	OECD AI Principles, ENISA TL
	Tampering	Auto-responses issue unsafe physical commands to robots/IoT devices.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	MITRE ATT&CK, ENISA TL
DF.1.2.2 File Data (at rest / in processing)	Tampering	Malicious or biased samples inserted into stored files contaminate downstream stages.	Apply access control, encryption, versioning, and integrity hashes to storage; forbid public buckets and unauthorized replication.	MITRE ATLAS, ISO/IEC 23894, ENISA AI Threat Landscape, SLSA
	Info Disclosure	Stored files are copied or exfiltrated without authorization.	Apply access control, encryption, versioning, and integrity hashes to storage; forbid public buckets and unauthorized replication.	ISO/IEC 27701, ISO/IEC 27040, GDPR
	Tampering	Deleting, tampering with, or skewing file versions distorts evidence.	Ensure integrity of data, models, and pipelines via hashes, signatures, and a chain of trust.	OWASP Top 10 (Web), SLSA, ISO/IEC 27001
	DoS	Malicious encryption or deletion of storage causes loss of availability.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	ENISA TL, ISO/IEC 27040
DF.1.2.4 Task Automation Requests	Tampering	Malicious instructions embedded in automation request bodies contaminate the chain.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	MITRE ATLAS, OWASP LLM Top 10
	Tampering	Safety-filter-bypass inputs trigger harmful actions.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	OWASP LLM Top 10, ISO/IEC 23894
	DoS	Bulk requests overload the automation pipeline.	Apply rate-limiting, backoff, and bot-blocking to crawling and bulk requests to prevent outages.	OWASP API Security Top 10, ENISA TL
DF.1.2.6 Configuration Data	Info Disclosure	Other users' data is extracted during preprocessing.	Enforce tenant isolation and schema/content validation in preprocessing and retriever stages.	ISO/IEC 27701, GDPR, OWASP LLM Top 10
	Info Disclosure	Misconfiguration causes personal data to appear in outputs.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	ISO/IEC 27701, GDPR, ISO/IEC 23894
	Info Disclosure	Missing query limits or rate limits enable model extraction and sensitive-data exposure.	Apply DLP, masking, and blocking policies so sensitive data is not included in outputs, logs, responses, or external transmissions.	Shokri et al. (CCS'17), OWASP LLM Top 10
	Tampering	Tampering with templates or system prompts contaminates instructions.	Manage system prompts/templates as immutable resources; control changes with dual approval and audit.	MITRE ATLAS, OWASP LLM Top 10
	Elevation of Privilege	Plugin or tool misconfiguration enables code execution.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	CWE-78/94/502, OWASP Top 10 (Web)
	Repudiation	Missing or altered logs prevent activity tracing.	Maintain tamper-evident audit logs with central collection, signing, and integrity verification; control access and changes.	NIST SP 800-92, ASVS
	Tampering	Incorrect dependencies/paths allow malicious package injection.	Issue short-lived access tokens with minimum scope and enable automatic rotation.	SLSA v1.0, OWASP Supply-Chain Guidance
	Tampering	Policy-violating configuration permits unauthorized actions.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	OWASP ASVS, OWASP LLM Top 10
	Tampering	Misconfiguration produces overconfident, erroneous outputs.	Use uncertainty estimation, warning badges, and source attribution to reduce overconfident outputs and edge-case errors.	ISO/IEC 23894, ENISA TL
	Tampering	Missing refusal messages leads to acceptance of dangerous requests.	Implement standardized refusal logic and templates for high-risk requests in a separate engine to prevent bypass.	OWASP LLM Top 10
	Elevation of Privilege	Poor exception/boundary handling causes malfunctions.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	ASVS, ISO/IEC 23894
	Tampering	Reward-system changes reinforce risky behaviors.	When changing reward systems, require offline validation, simulation, and approval to avoid reinforcing risky behaviors.	Christiano et al. (NeurIPS'17), Ouyang et al. (NeurIPS'22)
DF.8.1 Labeled Data	Tampering	Malicious or biased samples are inserted into labeled tuning sets.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	MITRE ATLAS, ISO/IEC 23894, ENISA AI Threat Landscape
	Tampering	Label flipping alters intended behavior.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	ISO/IEC TR 24027, ISO/IEC 23894
	Tampering	A trigger-response backdoor is inserted during tuning.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	MITRE ATLAS, ENISA TL
	Info Disclosure	Sensitive data leaks from labeled datasets.	Apply DLP, masking, and blocking policies so sensitive data is not included in outputs, logs, responses, or external transmissions.	ISO/IEC 27701, GDPR, ISO/IEC 27040
	Tampering	Label quality degradation causes performance erosion and drift.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	ISO/IEC 23894, ENISA TL
	Info Disclosure	Label schemas/structures are exposed externally.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	OWASP ASVS, OWASP API Security Top 10
DF.8.2 Training Data	Tampering	Malicious samples are injected into the training data.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	MITRE ATLAS, ISO/IEC 23894
	Tampering	Biased data is mixed into the dataset.	Establish procedures to measure and mitigate bias in inputs, datasets, and model outputs to ensure fairness.	ISO/IEC TR 24027, ENISA TL
	Info Disclosure	Datasets are copied without authorization or exposed via public buckets.	Apply access control, encryption, versioning, and integrity hashes to storage; forbid public buckets and unauthorized replication.	ISO/IEC 27701, ISO/IEC 27040, OWASP Cloud/Storage
	Repudiation	Removing watermarks leads to loss of traceability.	Apply watermarking/fingerprinting to critical data and alert/investigate upon removal.	ENISA TL, ISO/IEC 23894
	DoS	Malicious storage encryption or deletion causes loss of availability.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	ISO/IEC 27040, ENISA TL
	Tampering	Version/transform tampering breaks data integrity.	Ensure integrity of data, models, and pipelines via hashes, signatures, and a chain of trust.	SLSA, OWASP Top 10 (Web)
	Info Disclosure	Collection paths and bucket locations are exposed.	Apply access control, encryption, versioning, and integrity hashes to storage; forbid public buckets and unauthorized replication.	OWASP API Security Top 10, ASVS
	Spoofing	Storage access keys are exposed or stolen.	Apply access control, encryption, versioning, and integrity hashes to storage; forbid public buckets and unauthorized replication.	NIST SP 800-63B, OWASP ASVS
DF.1.2.3 User Input (Pre-processed Request, Contextual Information)	Tampering	Inputs contain malicious instructions before preprocessing.	Enforce tenant isolation and schema/content validation in preprocessing and retriever stages.	MITRE ATLAS, OWASP LLM Top 10
	Info Disclosure	Other users' data is extracted from preprocessed context.	Enforce tenant isolation and schema/content validation in preprocessing and retriever stages.	ISO/IEC 27701, GDPR
	DoS	Large query volumes overload preprocessing or the retriever.	Enforce tenant isolation and schema/content validation in preprocessing and retriever stages.	OWASP API Security Top 10, ENISA TL
	Rc(정찰)	Social-engineering queries elicit internal information.	Apply a standard non-disclosure response policy to internal-info requests to neutralize social-engineering queries.	ENISA TL, OWASP LLM Top 10
	Tampering	Malicious samples are planted in user-provided data sources.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	MITRE ATLAS, ISO/IEC 23894
	Tampering	Injected data bias induces model bias.	Establish procedures to measure and mitigate bias in inputs, datasets, and model outputs to ensure fairness.	ISO/IEC TR 24027, ISO/IEC 23894
DF.1.2.5 System Prompt	Tampering	Tampered system-prompt templates corrupt policies and behavior.	Manage system prompts/templates as immutable resources; control changes with dual approval and audit.	MITRE ATLAS, OWASP LLM Top 10
	Info Disclosure	System-prompt misconfiguration reveals sensitive information.	Apply DLP, masking, and blocking policies so sensitive data is not included in outputs, logs, responses, or external transmissions.	ISO/IEC 27701, GDPR, ISO/IEC 23894
	Tampering	The system's refusal logic is disabled.	Implement standardized refusal logic and templates for high-risk requests in a separate engine to prevent bypass.	OWASP LLM Top 10, ISO/IEC 23894
DF.1.2.7 Configured Settings	Info Disclosure	Workflow or session settings expose model versions or AI usage.	Minimize workflow/agent privileges, and execute high-risk commands only after human-in-the-loop approval.	OECD AI Principles, ENISA TL
	Tampering	Misconfiguration results in overconfident outputs and edge-case mistakes.	Use uncertainty estimation, warning badges, and source attribution to reduce overconfident outputs and edge-case errors.	ISO/IEC 23894, ENISA TL
	Tampering	Misconfigured plugin/tool privileges enable unsafe physical control.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	MITRE ATT&CK/ATLAS, ENISA TL
	Tampering	Agent settings/permissions cause goal misalignment or collusion.	Minimize workflow/agent privileges, and execute high-risk commands only after human-in-the-loop approval.	ISO/IEC 23894, ENISA TL, OWASP LLM Top 10
DF.15.1 Prompt (Command, Context Information)	Tampering	Direct/indirect prompt injection into commands or context contaminates system instructions and bypasses policies.	The system must implement prompt-injection detection/blocking for all inputs and contexts and preserve system instructions.	MITRE ATLAS™ (Adversarial Threat Landscape for AI Systems), OWASP Top 10 for Large Language Model Applications / OWASP GenAI Security Project, ISO/IEC 23894:2023 — Information technology — Artificial intelligence — Guidance on risk management, ENISA Threat Landscape 2023, OWASP Application Security Verification Standard (ASVS) v4.0.3
	Info Disclosure	Injection forces the system prompt or PII to appear in output.	The system must implement prompt-injection detection/blocking for all inputs and contexts and preserve system instructions.	ISO/IEC 27701:2019 — Privacy Information Management System (PIMS), General Data Protection Regulation (EU) 2016/679 — GDPR, MITRE ATLAS™, OWASP Top 10 for Large Language Model Applications
	DoS	Repeated/parallel/chain queries exhaust compute, add latency, and spike billing.	The system must enforce rate limits and caps on retries/concurrency per session/user/API key to prevent resource exhaustion and excessive billing.	OWASP API Security Top 10 (2023), OWASP Top 10 for Large Language Model Applications, ENISA Threat Landscape 2023
	Elevation of Privilege	Prompts abuse plugin/tool privileges to execute commands or pivot laterally.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	OWASP Top 10 for Large Language Model Applications, Common Weakness Enumeration CWE-502 — Deserialization of Untrusted Data, CWE-78 — OS Command Injection, CWE-94 — Improper Control of Generation of Code (Code Injection)
DF.16.1 Output (Status Code, Error Messages)	Tampering	Stealth code/links/QRs are inserted into status or error outputs.	Allow only safe markup for rendering outputs and statically analyze to block injected code, links, or QR content.	OWASP Top 10 for Large Language Model Applications, CWE-79 — Improper Neutralization of Input During Web Page Generation (Cross-site Scripting), CWE-94 — Improper Control of Generation of Code
	Info Disclosure	Error messages expose fabricated evidence or sensitive data, which then spreads.	Apply DLP, masking, and blocking policies so sensitive data is not included in outputs, logs, responses, or external transmissions.	ISO/IEC 23894:2023, ENISA Threat Landscape 2023, OWASP Top 10 for Large Language Model Applications
DF.15.2 Web API	Tampering	Hidden instructions in API responses automate ToS-violating calls.	Strip hidden instructions from both structured and unstructured content (e.g., JSON/XML/headers, search docs).	OWASP Top 10 for Large Language Model Applications, OWASP API Security Top 10 (2023), ISO/IEC 23894:2023
	Info Disclosure	Plugins/endpoints exfiltrate sensitive data or expose schemas.	Apply DLP, masking, and blocking policies so sensitive data is not included in outputs, logs, responses, or external transmissions.	ISO/IEC 27701:2019, GDPR, OWASP API Security Top 10 (2023)
	DoS	Automation/crawling causes excessive calls, leading to latency, outages, and higher cost.	Apply rate-limiting, backoff, and bot-blocking to crawling and bulk requests to prevent outages.	OWASP API Security Top 10 (2023), ENISA Threat Landscape 2023
	Elevation of Privilege	Processing a response with a deserialization flaw enables remote code execution.	Avoid unsafe (de)serialization and enforce schema validation, timeouts, and size limits.	CWE-502 — Deserialization of Untrusted Data, OWASP Top 10 (Web Applications, 2021) A8 — Software and Data Integrity Failures
DF.16.2 HTTP Response (JSON, Error Messages)	Tampering	Hidden instructions in JSON/XML/headers contaminate downstream chains.	Strip hidden instructions from both structured and unstructured content (e.g., JSON/XML/headers, search docs).	OWASP Top 10 for Large Language Model Applications, OWASP API Security Top 10 (2023)
	Elevation of Privilege	Exploiting a deserialization bug leads to remote code execution.	Avoid unsafe (de)serialization and enforce schema validation, timeouts, and size limits.	CWE-502 — Deserialization of Untrusted Data, OWASP Top 10 (Web Applications, 2021) A8 — Software and Data Integrity Failures
	Info Disclosure	Malicious links or scripts are injected into responses.	Allow only safe markup for rendering outputs and statically analyze to block injected code, links, or QR content.	OWASP Top 10 for Large Language Model Applications, CWE-79 / CWE-94
DF.6.3.1 Prompt (Next-step Instructions, Loop data)	DoS	Infinite loops or high-cost steps exhaust resources and spike billing.	The system must enforce rate limits and caps on retries/concurrency per session/user/API key to prevent resource exhaustion and excessive billing.	OWASP Top 10 for Large Language Model Applications, ENISA Threat Landscape 2023
	Tampering	Reward-model weaknesses reinforce dangerous next steps.	When changing reward systems, require offline validation, simulation, and approval to avoid reinforcing risky behaviors.	Paul Christiano et al., "Deep Reinforcement Learning from Human Preferences," NeurIPS 2017, Ouyang et al., "Training language models to follow instructions with human feedback (InstructGPT)," NeurIPS 2022, ISO/IEC 23894:2023
DF.7.1.1 Response (Generated Text, Function	Tampering	Stealth code/links are inserted into generated outputs or function parameters.	Allow only safe markup for rendering outputs and statically analyze to block injected code, links, or QR content.	OWASP Top 10 for Large Language Model Applications, CWE-79 / CWE-94
	Info Disclosure	Suppressing or bypassing refusals leads to harmful/illegal content generation.	Implement standardized refusal logic and templates for high-risk requests in a separate engine to prevent bypass.	OWASP Top 10 for Large Language Model Applications, ISO/IEC 23894:2023
DF.4.1 API/Command Calls (Function calls)	Tampering	Abusing function/tool call chains enables command injection or code execution.	The system must implement prompt-injection detection/blocking for all inputs and contexts and preserve system instructions.	CWE-78 — OS Command Injection, CWE-502 — Deserialization of Untrusted Data, OWASP Top 10 for Large Language Model Applications
	Info Disclosure	Plugin/tool calls exfiltrate sensitive data and expose endpoints/schemas.	Apply DLP, masking, and blocking policies so sensitive data is not included in outputs, logs, responses, or external transmissions.	ISO/IEC 27701:2019, GDPR, OWASP API Security Top 10 (2023)
	DoS	Repeated/nested function calls exhaust CPU and memory and add latency.	The system must enforce rate limits and caps on retries/concurrency per session/user/API key to prevent resource exhaustion and excessive billing.	OWASP API Security Top 10 (2023), ENISA Threat Landscape 2023
	Tampering	Third-party APIs are called in violation of terms/policies.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	OWASP API Security Top 10 (2023), OWASP Top 10 for Large Language Model Applications
DF.4.2 HTTP Response (JSON, XML)	Tampering	Hidden instructions in JSON/XML/headers contaminate the chain.	Strip hidden instructions from both structured and unstructured content (e.g., JSON/XML/headers, search docs).	OWASP Top 10 for Large Language Model Applications, OWASP API Security Top 10 (2023)
	Elevation of Privilege	A response-deserialization flaw enables remote code execution.	Avoid unsafe (de)serialization and enforce schema validation, timeouts, and size limits.	CWE-502 — Deserialization of Untrusted Data, OWASP Top 10 (Web Applications, 2021) A8 — Software and Data Integrity Failures
	Tampering	Response bodies include malicious links or scripts.	Allow only safe markup for rendering outputs and statically analyze to block injected code, links, or QR content.	OWASP Top 10 for Large Language Model Applications, CWE-79 / CWE-94

Asset	STRIDE (Threat) / Safety Risk	Threat Scenario (EN)	Requirement (shall) (EN)	Evidence (Representative Docs)
DF.7.2.1 Fine Tuning Dataset (Domain-Specific texts, Labeled examples)	Tampering	Malicious/poisoned samples in tuning data alter model behavior.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	MITRE ATLAS™ (Adversarial Threat Landscape for AI Systems), ISO/IEC 23894:2023 — Information technology — Artificial intelligence — Guidance on risk management, ENISA Threat Landscape 2023, SLSA: Supply-chain Levels for Software Artifacts v1.0
	Tampering	Label flipping or omissions mislead training objectives.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	ISO/IEC TR 24027:2021 — Bias in AI systems and AI-aided decision making, ISO/IEC 23894:2023
	Tampering	A trigger-response backdoor is inserted during tuning.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	MITRE ATLAS™, ENISA Threat Landscape 2023
	Info Disclosure	PII/secret data in the tuning set leaks to outputs/logs.	Apply DLP, masking, and blocking policies so sensitive data is not included in outputs, logs, responses, or external transmissions.	ISO/IEC 27701:2019 — Privacy Information Management System, GDPR (Regulation (EU) 2016/679), ISO/IEC 27040:2015 — Storage security
	Tampering	Low-quality/imbalanced tuning causes performance erosion and drift.	For labels, tuning, and training data, apply provenance verification, PII removal, backdoor detection, and quality monitoring.	ISO/IEC 23894:2023, ENISA Threat Landscape 2023
DF.7.2.2 Large-Scale Raw Dataset	Tampering	Toxic/biased text is mixed into the large raw corpus.	Establish procedures to measure and mitigate bias in inputs, datasets, and model outputs to ensure fairness.	MITRE ATLAS™, ISO/IEC 23894:2023, ENISA Threat Landscape 2023
	Info Disclosure	Public buckets or mis-set permissions expose raw datasets.	Apply access control, encryption, versioning, and integrity hashes to storage; forbid public buckets and unauthorized replication.	ISO/IEC 27701:2019, ISO/IEC 27040:2015, OWASP API Security Top 10 (2023)
	DoS	Malicious storage encryption/deletion causes loss of availability.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	ENISA Threat Landscape 2023, ISO/IEC 27040:2015
	Tampering	Tampering with version/transform pipelines breaks integrity.	Ensure integrity of data, models, and pipelines via hashes, signatures, and a chain of trust.	SLSA v1.0, OWASP Top 10 (Web Applications, 2021) — A8: Software and Data Integrity Failures
DF.6.3.2 Dataset (Pipeline)	Tampering	Malicious batches in intermediate pipeline sets contaminate downstream stages.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	MITRE ATLAS™, ISO/IEC 23894:2023, SLSA v1.0
	Tampering	Altered data paths or cleansing rules force unintended processing.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	OWASP API Security Top 10 (2023), ENISA Threat Landscape 2023
	Info Disclosure	Intermediate data leaks via logs or third-party storage paths.	Maintain tamper-evident audit logs with central collection, signing, and integrity verification; control access and changes.	ISO/IEC 27701:2019, ISO/IEC 27040:2015
	DoS	Queue/ETL overload causes delays and stoppages.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat and avoid operational impact.	OWASP API Security Top 10 (2023), ENISA Threat Landscape 2023
	Tampering	External storage disagrees with or tampers with internal intermediates.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	ISO/IEC 27001:2022 — ISMS Requirements, SLSA v1.0
DF.7.3.1 Specialized Dataset (Domain-Specific Documents, Labeled Examples)	Tampering	Domain-specific sets with biased/low-quality samples amplify inference errors.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	ISO/IEC TR 24027:2021, ISO/IEC 23894:2023
	Info Disclosure	Restricted/sensitive domain documents are exposed via search or summarization.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ISO/IEC 27701:2019, GDPR
	DoS	Repeating domain-specific queries overloads the retriever or index.	Limit repeats/loops/excessive calls to prevent latency and resource exhaustion.	Microsoft Learn — Retrieval-Augmented Generation (RAG) Guidance, NVIDIA "RAG 101: Build LLM Apps with Retrieval Augmented Generation", OWASP API Security Top 10 (2023)
DF.7.3.2 Large-Scale Raw Dataset (Text Corpus,	Tampering	Toxic/false/biased pages flood the crawling corpus, degrading model quality.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	MITRE ATLAS™, ISO/IEC 23894:2023, ENISA Threat Landscape 2023
	Info Disclosure	PII/confidential content in the corpus leaks during inference or summarization.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ISO/IEC 27701:2019, GDPR, ISO/IEC 27040:2015
DF.7.3.3 Raw External Data (Crawled, Unstructured,	Tampering	Hidden instructions in external data contaminate the chain before validation.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP Top 10 for Large Language Model Applications, Microsoft Learn — RAG Guidance, NVIDIA "RAG 101"
	Tampering	Malicious/manipulated samples enter through external supply paths.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	MITRE ATLAS™, SLSA v1.0, ISO/IEC 23894:2023
DF.9.1 Publicly Available Datasets (Web-Crawled	Tampering	Poisoned/distorted public or open-source data corrupts training/tuning.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	MITRE ATLAS™, ISO/IEC 23894:2023
	Info Disclosure	Sensitive/personal data mixed in public sets leaks via inference/summaries.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ISO/IEC 27701:2019, GDPR, ENISA Threat Landscape 2023
DF.20 Latent Representation	Info Disclosure	Latent vectors/embeddings reveal training traits or PII by inversion.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	Reza Shokri et al., "Membership Inference Attacks Against Machine Learning Models," ACM CCS 2017, ISO/IEC 23894:2023, ISO/IEC 20889:2018 — Privacy enhancing data de-identification — Terminology and classification of techniques
	Tampering	Malicious vectors bias similarity search and inference.	Verify the integrity and provenance of vector/embedding data to prevent retrieval/inference bias.	Microsoft Learn — RAG Guidance, NVIDIA "RAG 101: Build LLM Apps with Retrieval Augmented Generation", MITRE ATLAS™
	Tampering	Disturbed latent relations cause semantic drift and misclassification.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	ISO/IEC 23894:2023, ENISA Threat Landscape 2023
DF.18.1 Prompt, Context	Tampering	Prompt injection in the context window contaminates system instructions and bypasses policies.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	MITRE ATLAS, OWASP LLM Top 10, ISO/IEC 23894, ENISA TL 2023, OWASP ASVS
DF.17.1 User Query	Tampering	Jailbreak/bypass prompts disable safeties and induce harmful output.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	OWASP LLM Top 10, ISO/IEC 23894
	Info Disclosure	Benign-looking queries spread false or manipulated information.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	ISO/IEC 23894, ENISA TL 2023
DF.21 Retrieved context	Tampering	Hidden instructions in retrieved documents/summaries contaminate the chain.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP LLM Top 10, Microsoft RAG Guidance, NVIDIA RAG 101, MITRE ATLAS
	Info Disclosure	RAG results include or expose unauthorized/sensitive documents.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ISO/IEC 27701, GDPR, ISO/IEC 27040, Microsoft RAG, ENISA TL 2023
DF.22 Vector Search	DoS	Surges of high-cost similarity queries cause search delays and billing spikes.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP API Sec Top 10, ENISA TL 2023, Microsoft RAG
	Tampering	Poisoned vector indexes/embeddings rank attack documents higher.	Verify the integrity and provenance of vector/embedding data to prevent retrieval/inference bias.	MITRE ATLAS, Microsoft RAG, NVIDIA RAG 101
	Tampering	Injected malicious vectors skew similarity and ranking.	Verify the integrity and provenance of vector/embedding data to prevent retrieval/inference bias.	Microsoft RAG, NVIDIA RAG 101, MITRE ATLAS
DF.23 Relevant Content	Tampering	Stealth code/links appear in a "related content" block.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP LLM Top 10, CWE-79/94
	Info Disclosure	Related content includes fake citations or misleading links.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	ISO/IEC 23894, ENISA TL 2023
DF.24 Vectorized Query	Info Disclosure	Query embeddings/parameters are reverse-inferred, exposing system traits or training properties.	Verify the integrity and provenance of vector/embedding data to prevent retrieval/inference bias.	Shokri MIA (CCS'17), ISO/IEC 23894, ISO/IEC 20889
	Tampering	Queries with adversarial vectors bias search or ranking.	Verify the integrity and provenance of vector/embedding data to prevent retrieval/inference bias.	Microsoft RAG, NVIDIA RAG 101, MITRE ATLAS
DF.27 Chunked Documents	Tampering	Hidden instructions/links in chunk bodies or meta fields contaminate the chain.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP LLM Top 10, Microsoft RAG, NVIDIA RAG 101, MITRE ATLAS
	Tampering	Chunk boundary/version/transform tampering distorts content and corrupts evidence.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	SLSA v1.0, OWASP Top 10 (Web) A8, ISO/IEC 27040
P.1.1.1 Encoder	Tampering	Mass-injected poisoned samples distort the encoder's vector space and downstream semantics.	Verify the integrity and provenance of vector/embedding data to prevent retrieval/inference bias.	ISO/IEC 23894 §7.4, ENISA AI Threat Landscape, MITRE ATLAS (Data Poisoning)
	Tampering	Label flips/noise gradually erode encoder quality, hurting translation/summarization.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	NeurIPS/ICML 정규 드리프트 연구, ISO/IEC 23894
	Info Disclosure	Embedding inversion/model inversion reconstructs training features (including PII).	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	Shokri et al., CCS'17; ISO/IEC 20889; MITRE ATLAS
	Info Disclosure	Mixed-in sensitive/confidential pages are learned by the encoder and later leak.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ISO/IEC 27701, GDPR 가이드, ENISA
	Spoofing	Steganographic/hidden commands in images/audio become misleading text signals after encoding.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	Carlini et al., USENIX'16; OWASP LLM A01
	Tampering	Biased data targeting specific groups harms fairness from the vector-representation stage.	Verify the integrity and provenance of vector/embedding data to prevent retrieval/inference bias.	ISO/IEC 24027, NIST AI RMF SP1270
	Tampering	Injecting hateful/harassing text into the corpus increases harmful expressions at generation.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	EU AI Act Annex III(2e), OECD AI Principles
P.1.1.2 Decoder	Information Disclosure	Published decoder-architecture flaws and weak parameters are identified and abused.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	MITRE CVE/NVD, ATLAS Vulnerability Scan
	Information Disclosure	Internal logic/configuration leaks via docs or code repositories.	Protect configuration changes/files with access control and integrity verification.	ENISA Threat Landscape, OWASP LLM A04
	Repudiation	API-pattern analysis and parameter tampering map the decoder's response structure.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP API Security Top10, MITRE Reconnaissance Techniques
	Elevation of Privilege	Decoder plugins or script paths allow arbitrary code execution.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	CWE-94, CWE-78, MITRE ATT&CK T1059
	Information Disclosure	Unintended internal debug/system messages appear in final outputs.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	CWE-209, ISO/IEC 27001 A.12.4
	Spoofing	Direct prompt injection in decoder inputs bypasses safety policies.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	OWASP LLM A01, MITRE ATLAS
	Tampering	The decoder is induced to output false or overconfident information.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	NIST AI RMF, EU AI Act Annex IV
	Tampering	After bypassing safeties, refusal messages are suppressed.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP LLM A02, ISO/IEC 23894
	Tampering	Impossible or erroneous commands are attempted as normal operations, causing abnormal behavior.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	ISO/IEC 25010, MITRE ATT&CK T1204
	Tampering	RLHF reward-model manipulation distorts decoder output patterns.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OpenAI Safety Research, NeurIPS RLHF Papers
	DoS	Queries that cause infinite loops or heavy computation exhaust decoder resources.	Limit repeats/loops/excessive calls to prevent latency and resource exhaustion.	MITRE ATT&CK T1499, ENISA AI DoS Guidance
P.1.2 Application Service	Information Disclosure	OpenAPI/Swagger/status endpoints and version info leak, revealing internal paths/schemas.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP API Security Top 10 (2023), OWASP ASVS v4.0.3, ENISA Threat Landscape 2023
	Repudiation	Automated scanners probe debug/unintended endpoints; audit trails are missing.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP Testing Guide, NIST SP 800-92
	Elevation of Privilege	Over-permissive roles/scopes enable administrator-level privilege escalation.	Protect configuration changes/files with access control and integrity verification.	ISO/IEC 27001:2022, OWASP ASVS
	Tampering	Exposed debug/admin features allow unauthorized function calls.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP ASVS, OWASP LLM Top 10
	Tampering	Config paths are abused to swap in malicious samples/models.	Protect configuration changes/files with access control and integrity verification.	MITRE ATLAS, ISO/IEC 23894:2023, SLSA v1.0
	Elevation of Privilege	Insiders change settings to overexpose data/model scope.	Protect configuration changes/files with access control and integrity verification.	ISO/IEC 27001:2022, NIST SP 800-53
	Information Disclosure	Tokens/keys are hard-coded in config files or environment variables and then leak.	Protect configuration changes/files with access control and integrity verification.	ISO/IEC 27701:2019, GitHub Security Advisory
	Information Disclosure	Output-format mistakes include personal or sensitive data in responses.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ISO/IEC 27701:2019, ISO/IEC 23894:2023
	Information Disclosure	Abusing rate-limit/query-width allows model/knowledge extraction.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	Shokri et al., CCS'17, OWASP LLM Top 10
	Tampering	Tampering with system/chain templates bypasses policies and corrupts behavior.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP LLM Top 10, MITRE ATLAS
	Tampering	Weak plugin/tool configuration and checks enable remote code execution.	Protect configuration changes/files with access control and integrity verification.	CWE-502/78/94, OWASP Top 10 (Web, 2021)
	Repudiation	Poor activity/result logging and auditing make tracing impossible.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	NIST SP 800-92, ISO/IEC 27001
	Tampering	Dependency hijacking/registry tampering injects malicious packages.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	SLSA v1.0, OWASP Software Supply Chain Security
	Tampering	Calls automate regional/charging/terms bypass.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP API Security Top 10 (2023), ISO/IEC 23894:2023
	Tampering	Output settings increase overconfident or hallucinatory responses.	Protect configuration changes/files with access control and integrity verification.	ISO/IEC 23894:2023, ENISA TL 2023
	Tampering	Missing refusal messages or safety filters lead to dangerous requests being served.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP LLM Top 10
	Tampering	Vague or impossible commands are guessed and executed, causing malfunctions.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP LLM Top 10, ASVS
	Tampering	Reward-system bias and edge cases reinforce dangerous behavior.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	Christiano et al., NeurIPS'17, Ouyang et al., NeurIPS'22
	Information Disclosure	Tokens or session IDs leak via headers, logs, or redirect paths.	Filter logs to avoid recording model versions, prompts, and sensitive information.	NIST SP 800-63B, ISO/IEC 27701:2019
	Tampering	Intercepted callbacks/webhooks/redirects allow parameter tampering.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP API Security Top 10 (2023)
	Denial of Service	Request storms, loops, or excessive calls exhaust resources and spike billing.	Limit repeats/loops/excessive calls to prevent latency and resource exhaustion.	OWASP API Security Top 10 (2023), ENISA TL 2023
	Information Disclosure	Plugins or external APIs automatically transmit PII/confidential data.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ISO/IEC 27701:2019, GDPR
	Information Disclosure	Logs/banners/responses expose model versions or the system prompt.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	OECD AI Principles, ENISA TL 2023
	Tampering	Remote code execution occurs via a deserialization-vulnerable path.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	CWE-502, OWASP Top 10 (Web, 2021) A8

Asset	STRIDE (Threat) / Safety Risk	Threat Scenario (EN)	Requirement (shall) (EN)	Evidence (Representative Docs)
P.1.2.1 Multi-Modal	Information Disclosure	Multimodal processing leaks decoding-pipeline or model-structure details.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	OWASP LLM A04, MITRE ATLAS Reconnaissance
	Spoofing	Credentials (API keys, auth tokens) are stolen through multimodal input paths.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	CWE-522, OWASP API Security
	Information Disclosure	Multimodal responses include confidential images or audio.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	ENISA AI Threat Landscape, NIST SP 800-53 SC-28
	Information Disclosure	Mixed image/audio/text responses expose inappropriate content.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	ISO/IEC 23894, OWASP LLM A05
	Tampering	Multimodal datasets are poisoned during training/tuning.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	MITRE ATLAS T0050, IEEE AI Poisoning Survey
	Tampering	Image/audio metadata is used for indirect prompt injection.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	OWASP LLM A02, MITRE ATT&CK T1564
	Tampering	Adversarial patterns in image/audio inputs cause misclassification.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	NIPS AdvML, MITRE ATLAS
	Tampering	Persistent malicious inputs degrade multimodal performance.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	NIST AI RMF, ISO/IEC 24029-1
	DoS	Large image/audio uploads cause delays and denial of service.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	MITRE ATT&CK T1499, ENISA AI DoS Guidance
	Tampering	Image-/audio-based prompt injection bypasses safety policies.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	OWASP LLM A02, MITRE ATLAS
	Tampering	Multimodal plugin features are abused to execute system commands.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	CWE-78, MITRE ATT&CK T1203
	Information Disclosure	Multimodal outputs contain hidden or malicious code/links.	Detect and block hidden instructions and adversarial patterns in multimodal inputs/outputs.	CWE-94, ISO/IEC 27001 A.12.2
P.1.2.2 Workflow System	Information Disclosure	Exposed workflow design/business logic enables policy bypass.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	ENISA AI TL, ISO/IEC 27005
	Tampering	Malicious workflow requests are processed as valid, causing harm.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	OWASP LLM A05, MITRE ATLAS
	Tampering	Automated workflows mass-send hateful or harassing messages.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	EU AI Act Annex III, ISO 37106
	Tampering	Biased data/rules propagate through all workflow stages.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	NIST AI RMF, ISO/IEC 24027
	Tampering	Domain-specific workflow errors repeat and amplify.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	ENISA AI TL, IEEE Access AI Safety
	Information Disclosure	Workflow logs or responses reveal AI usage or model versions.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	ISO/IEC 42001, OWASP LLM A09
	Tampering	Explanations during reasoning contain false information.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	NIST AI RMF, ISO/IEC 24027
	Tampering	Overconfidence controls for automated workflows are insufficient.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	EU AI Act Art.14, ISO 26262
	Tampering	Missing boundary/exception handling produces dangerous outcomes.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	ENISA AI TL, NIST SP 800-30
	Tampering	IoT/robot-control workflows are abused to manipulate the physical world.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	MITRE ATT&CK T0851, ISO 10218
	Tampering	Multi-agent workflows suffer goal misalignment.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	IEEE Spectrum AI Risks, ISO/IEC 42001
	Tampering	Agents collude within a workflow to bypass policies.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	MITRE ATLAS, ENISA AI TL
	Tampering	Long-term workflow memory is abused.	Minimize workflow/agent privileges and require pre-execution validation for risky commands.	OWASP LLM A06, ISO/IEC 27040
P.1.2.3 LLM App Configuration	Information Disclosure	Config files expose known vulnerabilities or settings.	Protect configuration changes/files with access control and integrity verification.	MITRE CVE DB, NIST NVD
	Information Disclosure	LLM app settings (tokens, API endpoints) are exposed.	Protect configuration changes/files with access control and integrity verification.	OWASP LLM A03, GitHub Advisory
	Repudiation	Automated scanners uncover configuration weaknesses.	Protect configuration changes/files with access control and integrity verification.	ENISA AI TL, OWASP Testing Guide
	Elevation of Privilege	Misconfiguration grants administrator-level privileges.	Protect configuration changes/files with access control and integrity verification.	ISO/IEC 27001 A.9
	Tampering	Exposed debug/experimental features allow unauthorized access.	Implement required security controls—validation, filtering, integrity checks, and monitoring—to mitigate the threat.	OWASP LLM A04
	Tampering	Settings are altered to inject model backdoors.	Protect configuration changes/files with access control and integrity verification.	MITRE ATLAS T1584
	Elevation of Privilege	Insiders change configurations to gain full model access.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	ISO 27035-1, NIST SP 800-53
	Information Disclosure	Sensitive data is stored inside configuration files.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	NIST 800-171, OWASP Top 10
	Information Disclosure	Output settings inject personal data into responses.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ISO/IEC 27559, ENISA AI TL
	Information Disclosure	Missing query-limit settings allow model extraction.	Limit repeats/loops/excessive calls to prevent latency and resource exhaustion.	MITRE ATLAS T1606
	Tampering	Tampered prompt-template settings manipulate the model.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A01
	Tampering	Config errors let external plugins be abused for code execution.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	MITRE ATT&CK T1203
	Repudiation	Missing/altered logging and audit trails prevent tracing.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	ISO/IEC 27037
	Tampering	Dependency hijacking or registry tampering injects malicious packages.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	SLSA, OWASP Supply Chain
	Tampering	Calls automate region/charging/terms bypass.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	ISO 27001 A.5
	Tampering	Output settings increase overconfident or hallucinatory replies.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	NIST AI RMF
	Tampering	Missing refusal messages lead to dangerous requests being answered.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	EU AI Act Art. 14
	Tampering	Faulty command-handling logic causes errors.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	OWASP LLM A10
	Tampering	Reward-system changes induce unintended learning.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OpenAI System Card, Anthropic Safety Report
P.1.3 LLM Application (Agents)	Information Disclosure	Agent architecture and version information are exposed.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	OWASP LLM A05
	Spoofing	Attempts are made to steal API keys and account credentials.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	NIST SP 800-63B
	Information Disclosure	Sensitive data leaks during agent-to-agent communications.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	ISO/IEC 27002 A.8
	Information Disclosure	Outputs are risky or insufficiently validated.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A06
	Tampering	Inappropriate training data is injected into long-term memory.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	MITRE ATLAS T1601
	Tampering	Indirect prompt injection enters via external data.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A01
	Tampering	Adversarial image/audio prompts cause malfunctions.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	ENISA AI TL
	Denial of Service	Repetitive tasks degrade model performance.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	MITRE ATLAS T1499
	Denial of Service	Multi-agent overload causes denial of service.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A09
	Tampering	Hidden instructions in agent-handled image/audio inputs.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A01
	Tampering	External tools/plugins are misused.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	MITRE ATT&CK T1203
	Information Disclosure	Malicious code or harmful content is generated.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	OWASP LLM A08
	Information Disclosure	Data/model bias harms specific groups.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	IEEE P7003
	Information Disclosure	Errors or misinformation appear in specialized domains (e.g., medical, legal).	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	NIST AI RMF
	Spoofing	The agent conceals that it is AI and imitates a human.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	EU AI Act Art. 52
	Information Disclosure	No explanation is provided for the reasoning process.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	ISO/IEC 24027
	Tampering	Human-oversight procedures are bypassed.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	EU AI Act Art. 14
	Tampering	Poor task definitions cause edge-case errors.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ENISA AI TL
	Tampering	Errors occur in physical control of IoT/robotics.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	ISO 10218-2
	Tampering	Agent goals conflict with system intent.	Verify the integrity and provenance of embeddings/vectors/encoders and use outlier detection to prevent contamination.	Anthropic Safety Research
	Denial of Service	Infinite loops or excessive resource usage occur.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	MITRE ATT&CK T1499
	Tampering	Multi-agent collaboration produces harmful outcomes.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	ENISA AI TL
	Information Disclosure	Long-term memory data is abused.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	OWASP LLM A07
P.1.4.1 Plugin - OS Process	Elevation of Privilege	A malicious plugin loads with OS-level privileges and accesses system resources.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	MITRE ATT&CK T1203, OWASP LLM A07
	Tampering	The plugin executes arbitrary shell commands, altering the local system.	Minimize plugin/tool privileges, validate in a sandbox before release, and monitor external calls.	CWE-78, CWE-94
	Info Disclosure	API keys/tokens are stolen from environment variables or the keychain.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP ASVS, NIST SP 800-63B
	Tampering	Tampering with config/log files evades detection.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	NIST SP 800-92
	DoS	Infinite loops/high I/O exhaust CPU and memory.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A09
	Tampering	A backdoor is planted via the plugin update path.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	SLSA, OWASP Supply Chain Top10
	Info Disclosure	Sensitive data is exfiltrated from local files/cache.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	ISO/IEC 27040
P.1.4.2 Plugin - Web based	Tampering	External API traffic is intercepted and responses are tampered with.	When changing reward models, validate and monitor to detect any reinforcement of risky behaviors.	OWASP API Sec, MITM Guides
	Tampering	Hidden prompts are embedded in external responses.	Limit repeats/loops/excessive calls to prevent latency and resource exhaustion.	OWASP LLM A01
	Info Disclosure	Plugin calls exfiltrate user PII.	Minimize plugin/tool privileges, validate in a sandbox before release, and monitor external calls.	ISO/IEC 27701
	Tampering	A plugin-response deserialization flaw enables remote code execution.	Implement immutable audit logs and trigger alerts and response procedures on tampering.	CWE-502, OWASP Top10 A8
	Tampering	Third-party APIs are called in violation of terms/policies.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	ISO/IEC 23894
	Info Disclosure	Plugin endpoints/schemas are public and abused.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A04
	Tampering	Malicious links/scripts are inserted into model outputs.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	CWE-79/94
P.1.5.1 Generation Module	Tampering	Stealth commands/code/QRs are embedded in generated results.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	OWASP LLM A08
	Tampering	Prompts induce violent/self-harm/terror-related content.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	OWASP LLM A06
	Tampering	False or manipulated claims are mass-generated.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	UNESCO/EU AI Act
	Tampering	Overconfident but incorrect explanations/diagnoses are provided.	Apply rate limits and query-complexity caps to prevent model or knowledge extraction.	NIST AI RMF
	Tampering	Refusal suppression handles dangerous requests.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A01
	Tampering	Undefined requirement boundaries/exceptions are exploited.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	ENISA AI TL
	Tampering	Reward-model weaknesses increase harmful outputs.	Implement immutable audit logs and trigger alerts and response procedures on tampering.	RLHF 연구(NeurIPS/ICML)
P.1.5.2 Retrieval Module	Tampering	Malicious embeddings in the vector index make poisoned documents rank first.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	NVIDIA RAG 101, MITRE ATLAS
	Tampering	Hidden prompts in retrieved documents are executed.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP LLM A01
	Info Disclosure	RAG indexing scope/policy leaks, enabling targeted attacks.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	MS/Azure RAG Guidance
	DoS	Surges of high-cost similarity queries delay service.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP API DoS
	Info Disclosure	Unauthorized documents appear in search/summarization results.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	ISO/IEC 27040
	Tampering	Inaccurate domain retrieval produces incorrect evidence.	When changing reward models, validate and monitor to detect any reinforcement of risky behaviors.	ENISA AI TL
P.1.5.3 Embedding Model	Tampering	Poisoned data corrupts the embedding space, boosting harmful retrievals.	Implement immutable audit logs and trigger alerts and response procedures on tampering.	MITRE ATLAS, NVIDIA RAG Security
	Tampering	Malicious vectors keyed to certain terms skew similarity.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	NIST AI RMF
	Info Disclosure	Embedding-model parameters are reverse-inferred, leaking IP.	Limit repeats/loops/excessive calls to prevent latency and resource exhaustion.	ISO/IEC 27001
	Tampering	Altered word-relation geometry induces wrong results.	Apply DLP and masking so sensitive data is not exposed in output, storage, or transmission.	ENISA AI TL
	Info Disclosure	Embedding values are exposed directly via API responses.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	OWASP LLM A04
	Tampering	Training-data bias in embeddings spreads to search/recommendation.	Eliminate deserialization vulnerabilities and use safe parsers and validation for external inputs.	IEEE P7003
P.2 Vector DB Storage	Tampering	Malicious vectors injected into a DB index distort results.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	MITRE ATLAS
	Info Disclosure	Unauthorized users search embeddings of sensitive documents.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	ISO/IEC 27040
	DoS	Large similarity-query volumes overload CPU and memory.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	OWASP API Security

Asset	STRIDE (Threat) / Safety Risk	Threat Scenario (EN)	Requirement (shall) (EN)	Evidence (Representative Docs)
D.1 Vector DB Storage	Tampering	Portions of vector data are deleted or tampered with.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	ISO/IEC 27001
	Info Disclosure	Index metadata (tags/locations) leaks externally.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	OWASP LLM A04
	Tampering	Connected plugins provide unauthorized data access.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	OWASP API Sec
D.1.1 Fine Tuning Data Storage	Tampering	Fine-tuning data includes malicious or biased samples.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	OWASP LLM A02
	Info Disclosure	Sensitive data leaks from the tuning dataset.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	ISO/IEC 27701
	Tampering	Label changes alter model behavior.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	IEEE P7003
	Tampering	During tuning, specific triggers produce malicious responses.	Implement security mechanisms that detect/block prompt injection and prevent policy subversion.	MITRE ATLAS
	Tampering	Post-tuning, baseline performance drops, raising false positives and misses.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	NIST AI RMF
	Info Disclosure	Tuning-data schemas/structures are exposed.	Detect/block adversarial patterns and hidden instructions in multimodal I/O and limit upload sizes.	OWASP LLM A04
D.1.2 Training Data Storage	Tampering	Malicious data is injected into the training set.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	OWASP LLM A02
	Info Disclosure	The full training set is copied or exfiltrated.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	ISO/IEC 27040
	Tampering	Large amounts of group-biased data are injected.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	IEEE P7003
	Tampering	Removing training-data watermarks eliminates traceability.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	NIST SP 800-92
	Info Disclosure	Data provenance/collection paths are exposed.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	OWASP LLM A04
	Tampering	Portions of source data are deleted or altered.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	ISO/IEC 27001
D.1.3 Cloud Storage	Info Disclosure	Missing access control exposes training/tuning data.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP Cloud Top 10
	DoS	Encrypted/deleted cloud-stored data is ransomed.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	MITRE ATT&CK
	Tampering	Cloud-object version tampering returns incorrect data.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	ISO/IEC 27001
	Info Disclosure	Cloud bucket names and path patterns are harvested.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	OWASP API Sec
	Tampering	Connected external apps access data without authorization.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	ENISA Cloud Sec
	Info Disclosure	Cloud-storage access keys are stolen.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	OWASP LLM A04
P.1.5.4 Data Pipeline	Tampering	Data is altered in intermediate pipeline stages.	Minimize workflow/agent privileges and require human review for high-risk tasks before execution.	ISO/IEC 27001
	DoS	Processing queues overload, causing delays and timeouts.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	OWASP API Sec
	Info Disclosure	Pipeline schemas/mapping rules are leaked.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	OWASP LLM A04
	Tampering	Malicious data batches distort model behavior.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	MITRE ATLAS
	Info Disclosure	Source data is exfiltrated before transformation.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	ISO/IEC 27040
	Tampering	Pipeline routing is changed, sending data to malicious stages.	Design and implement validation, filtering, access control, monitoring, and logging to mitigate the threat.	OWASP API Sec
E.1 External Data Storage	Info Disclosure	PII/sensitive data leaks from third-party databases.	Apply access control and integrity checks to settings/config files and enforce audit/approval for changes.	ISO/IEC 27701
	Tampering	External storage conflicts with internal data states.	Authenticate and encrypt communications with external storage/downstream services, and verify response integrity/provenance to prevent mismatch/tampering.	ISO/IEC 27001
	DoS	External-storage outages limit model functionality.	Authenticate and encrypt communications with external storage/downstream services, and verify response integrity/provenance to prevent mismatch/tampering.	ENISA AI TL
	Info Disclosure	External storage paths and infrastructure details are harvested.	Authenticate and encrypt communications with external storage/downstream services, and verify response integrity/provenance to prevent mismatch/tampering.	OWASP API Sec
	Tampering	Externally provided data contains malicious samples.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	MITRE ATLAS
	Info Disclosure	External-storage API credentials are stolen.	Authenticate and encrypt communications with external storage/downstream services, and verify response integrity/provenance to prevent mismatch/tampering.	OWASP LLM A04
E.2 World Wide Web	Info Disclosure	Web crawling is used to gather organization/model information.	Route crawler/collector content through a separate validation pipeline and control crawling traffic.	MITRE ATLAS
	Tampering	Model plugins connect users to malicious sites.	Enforce signature/integrity checks on plugin/app updates, run in sandboxes, and apply least-privilege.	OWASP LLM A06
	Tampering	Web-search results feed in malicious data.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	OWASP LLM A02
	Info Disclosure	Web-API responses expose sensitive information.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	OWASP API Sec
	DoS	Excessive web-crawling requests cause service outages.	Route crawler/collector content through a separate validation pipeline and control crawling traffic.	OWASP API Sec
	Info Disclosure	Web plugin versions/configuration details are collected.	Enforce signature/integrity checks on plugin/app updates, run in sandboxes, and apply least-privilege.	OWASP API Sec
E.3 External Downstream Service	Tampering	Responses forwarded to downstream services are tampered with.	Authenticate and encrypt communications with external storage/downstream services, and verify response integrity/provenance to prevent mismatch/tampering.	ISO/IEC 27001
	Info Disclosure	Sensitive data leaks through downstream service APIs.	Authenticate and encrypt communications with external storage/downstream services, and verify response integrity/provenance to prevent mismatch/tampering.	OWASP API Sec
	Tampering	External services execute malicious commands.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	MITRE ATT&CK
	Info Disclosure	Downstream endpoints/configuration details are harvested.	Authenticate and encrypt communications with external storage/downstream services, and verify response integrity/provenance to prevent mismatch/tampering.	OWASP API Sec
	Tampering	Malicious data propagates to downstream systems.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	OWASP LLM A05
	Tampering	Malicious code is injected into downstream dependencies.	Authenticate and encrypt communications with external storage/downstream services, and verify response integrity/provenance to prevent mismatch/tampering.	OWASP Supply Chain Sec
U.1 AI Users	Tampering	A user's malicious prompt alters model behavior.	Apply social-engineering filters to user input and logic to prevent prompt injection.	OWASP LLM A01
	Info Disclosure	Social-engineering tactics elicit internal information.	Apply social-engineering filters to user input and logic to prevent prompt injection.	MITRE ATLAS
	Tampering	Filter-bypass inputs generate harmful content.	Block requests to generate harmful/illegal content and require human review for suspicious outputs.	OWASP LLM A07
	Info Disclosure	Another user's data is extracted.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	ENISA AI TL
	DoS	Service is disrupted by mass query submissions.	Use rate limits, concurrency caps, and automatic blacklisting to prevent DoS and resource exhaustion.	OWASP API Sec
	Info Disclosure	Users are deanonymized via response-pattern analysis.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	OWASP AI Sec Guide
U.2 Data Provider	Tampering	Provided data contains malicious samples.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	MITRE ATLAS
	Tampering	Biased input data induces model bias.	Measure and monitor bias in data/models and design/apply bias-mitigation measures.	NIST AI RMF
	Info Disclosure	Personal data in supplied content is misused.	For personal data processing/transfer, apply encryption, anonymization, and access control, and honor data-minimization.	ISO/IEC 27701
	Info Disclosure	Adversaries gather types/structure of supplied data.	Restrict configuration and access so infra/endpoint details are not exposed externally; enable auto-alerts on exposure.	OWASP LLM A04
	Tampering	Malicious commands are embedded in data metadata.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	OWASP LLM A02
	Tampering	Supplier API keys are stolen and abused.	Secure external API credentials and apply least-privilege, short-lived tokens, and automatic rotation.	OWASP API Sec
U.3 AI Developer	Tampering	Model code commits include malware.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	OWASP Supply Chain Sec
	Tampering	Backdoors are inserted into model code or data.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	MITRE ATLAS
	Info Disclosure	Tooling versions and development-environment details are profiled.	Restrict configuration and access so infra/endpoint details are not exposed externally; enable auto-alerts on exposure.	OWASP API Sec
	Info Disclosure	Sensitive data leaks via debug ports or logs.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	OWASP API Sec
	Info Disclosure	AI source code is copied or exfiltrated without authorization.	For personal data processing/transfer, apply encryption, anonymization, and access control, and honor data-minimization.	OWASP LLM A04
	Tampering	Model hyperparameters are changed without authorization.	Control model-parameter changes and record signatures/verification/audit logs for any modification.	ENISA AI TL
U.4 AI Partner	Tampering	A partner misuses provided APIs.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	OWASP API Sec
	Info Disclosure	Shared datasets are copied or sold without permission.	Strictly control access to sensitive/proprietary assets and apply data-leak detection and anti-duplication controls.	ISO/IEC 27701
	Tampering	A partner distributes a malicious plugin.	Enforce signature/integrity checks on plugin/app updates, run in sandboxes, and apply least-privilege.	OWASP LLM A06
	Info Disclosure	A partner probes infrastructure and API endpoints.	Restrict configuration and access so infra/endpoint details are not exposed externally; enable auto-alerts on exposure.	OWASP API Sec
	Tampering	A partner's supplied data includes malicious samples.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	MITRE ATLAS
	Tampering	A partner uses AI in violation of license terms.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	ENISA AI TL
U.5 AI Application Provider	Tampering	App updates distribute malicious code.	Enforce signature/integrity checks on plugin/app updates, run in sandboxes, and apply least-privilege.	OWASP Supply Chain Sec
	Tampering	A clean model is swapped with a malicious one.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	MITRE ATLAS
	Info Disclosure	App-usage data is collected or leaked without consent.	Enforce signature/integrity checks on plugin/app updates, run in sandboxes, and apply least-privilege.	ISO/IEC 27701
	Tampering	Unauthorized features are enabled, violating policy.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	ENISA AI TL
	Info Disclosure	App internals and API endpoints are mapped.	Enforce signature/integrity checks on plugin/app updates, run in sandboxes, and apply least-privilege.	OWASP API Sec
	Info Disclosure	Exception handling exposes sensitive data.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	OWASP LLM A04
U.6 AI Producer	Tampering	Model parameters are tampered with during production.	Control model-parameter changes and record signatures/verification/audit logs for any modification.	OWASP Supply Chain Sec
	Tampering	Malicious data is injected during production.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	MITRE ATLAS
	Info Disclosure	A proprietary model in production is copied without authorization.	Strictly control access to sensitive/proprietary assets and apply data-leak detection and anti-duplication controls.	ISO/IEC 27040
	Tampering	A hidden backdoor is inserted before release.	Apply malicious sample detection, scanning, and quarantine for externally collected data/code.	MITRE ATLAS
	Info Disclosure	The production environment is scanned over the network.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	OWASP API Sec
	Tampering	AI compute hardware is accessed without authorization.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	ENISA AI TL
U.7 AI Customer	Tampering	Malicious prompts are submitted via feedback forms.	Apply social-engineering filters to user input and logic to prevent prompt injection.	OWASP LLM A01
	Info Disclosure	Probing queries map model capabilities and limits.	Design and implement validation, filtering, isolation, monitoring, and access control to mitigate the threat.	MITRE ATLAS
	DoS	Repetitive or excessive requests from customers disrupt service.	Use rate limits, concurrency caps, and automatic blacklisting to prevent DoS and resource exhaustion.	OWASP API Sec
	Info Disclosure	One customer's personal data is extracted by another.	For personal data processing/transfer, apply encryption, anonymization, and access control, and honor data-minimization.	ENISA AI TL
	Tampering	Generated images/text spread misinformation.	Block requests to generate harmful/illegal content and require human review for suspicious outputs.	NIST AI RMF
	Tampering	Users intentionally request illegal or harmful content.	Block requests to generate harmful/illegal content and require human review for suspicious outputs.	OWASP LLM A07